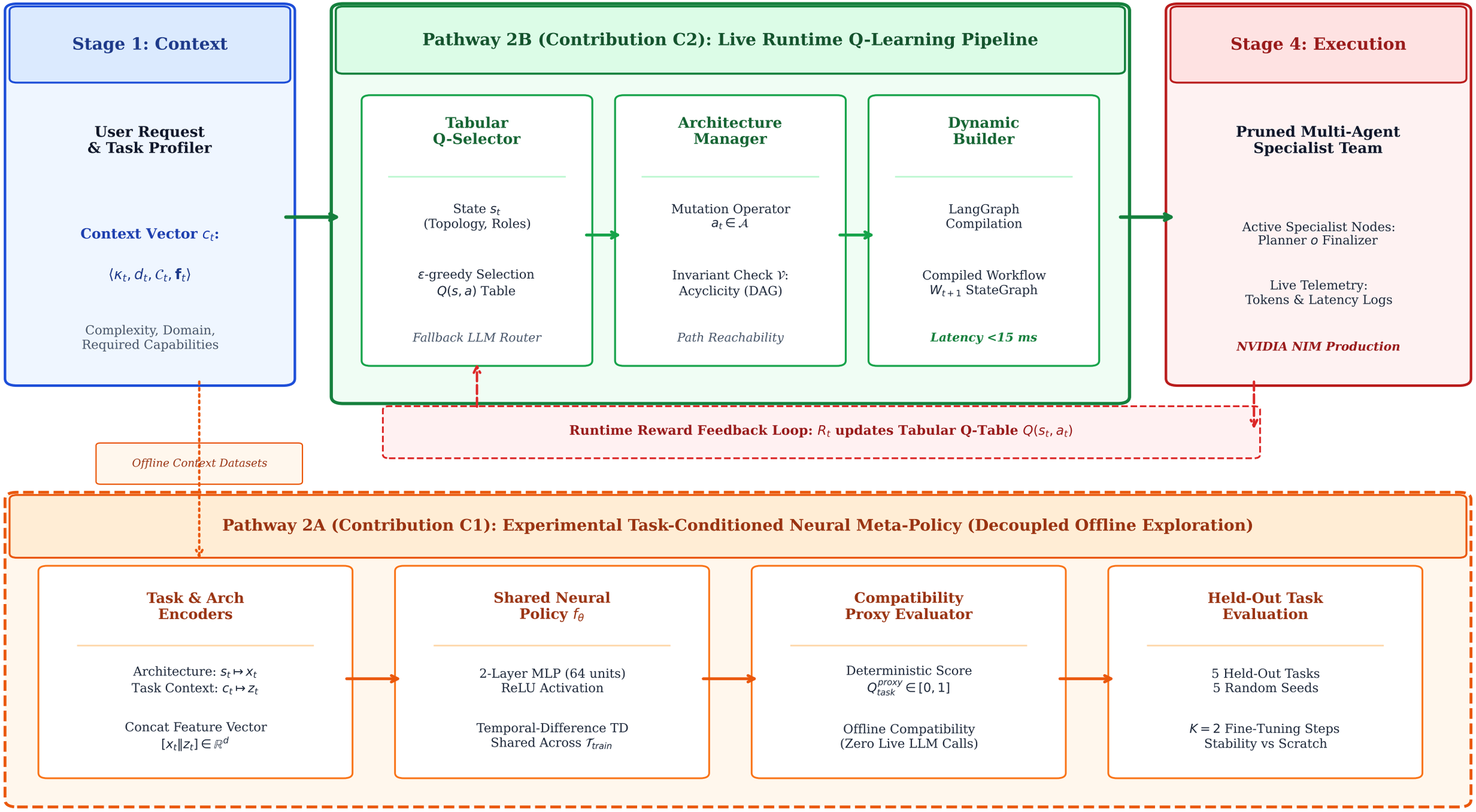

Stage 1: Context
User Request & Task Profiler
Context Vector c_t:
$\langle \kappa_t, d_t, C_t, \mathbf{f}_t \rangle$
Complexity, Domain, Required Capabilities
Pathway 2B (Contribution C2): Live Runtime Q-Learning Pipeline
Tabular Q-Selector
State s_t (Topology, Roles)
ε-greedy Selection $Q(s,a)$ Table
Fallback LLM Router
Architecture Manager
Mutation Operator $a_t \in \mathcal{A}$
Invariant Check $\mathcal{V}$: Acyclicity (DAG)
Path Reachability
Dynamic Builder
LangGraph Compilation
Compiled Workflow W_{t+1} StateGraph
Latency <15 ms
Stage 4: Execution
Pruned Multi-Agent Specialist Team
Active Specialist Nodes: Planner o Finalizer
Live Telemetry: Tokens & Latency Logs
NVIDIA NIM Production
Runtime Reward Feedback Loop: R_t updates Tabular Q-Table $Q(s_t, a_t)$
Offline Context Datasets
Pathway 2A (Contribution C1): Experimental Task-Conditioned Neural Meta-Policy (Decoupled Offline Exploration)
Task & Arch Encoders
Architecture: $s_t \mapsto x_t$
Task Context: $c_t \mapsto z_t$
Concat Feature Vector $[x_t \| z_t] \in \mathbb{R}^d$
Shared Neural Policy f_θ
2-Layer MLP (64 units) ReLU Activation
Temporal-Difference TD Shared Across $\mathcal{T}_{train}$
Compatibility Proxy Evaluator
Deterministic Score $Q_{task}^{proxy} \in [0,1]$
Offline Compatibility (Zero Live LLM Calls)
Held-Out Task Evaluation
5 Held-Out Tasks 5 Random Seeds
$K = 2$ Fine-Tuning Steps Stability vs Scratch